

Per-Position Mutual Information for MAC Polar Codes

Soft SC posteriors across channels and code classes

2026-04-13

Method

At each position i , the genie-aided SC decoder produces a posterior probability $p_i = P(U_i = 0 \mid Z^N, U_1^{i-1})$ (for user U) or $P(V_i = 0 \mid Z^N, V_1^{i-1})$ (for user V). The per-position MI is:

$$\text{MI}_i = 1 - H(p_i) = 1 - \mathbb{E}[-p_i \log_2 p_i - (1 - p_i) \log_2 (1 - p_i)]$$

This is the true mutual information of the i -th synthetic channel. It satisfies:

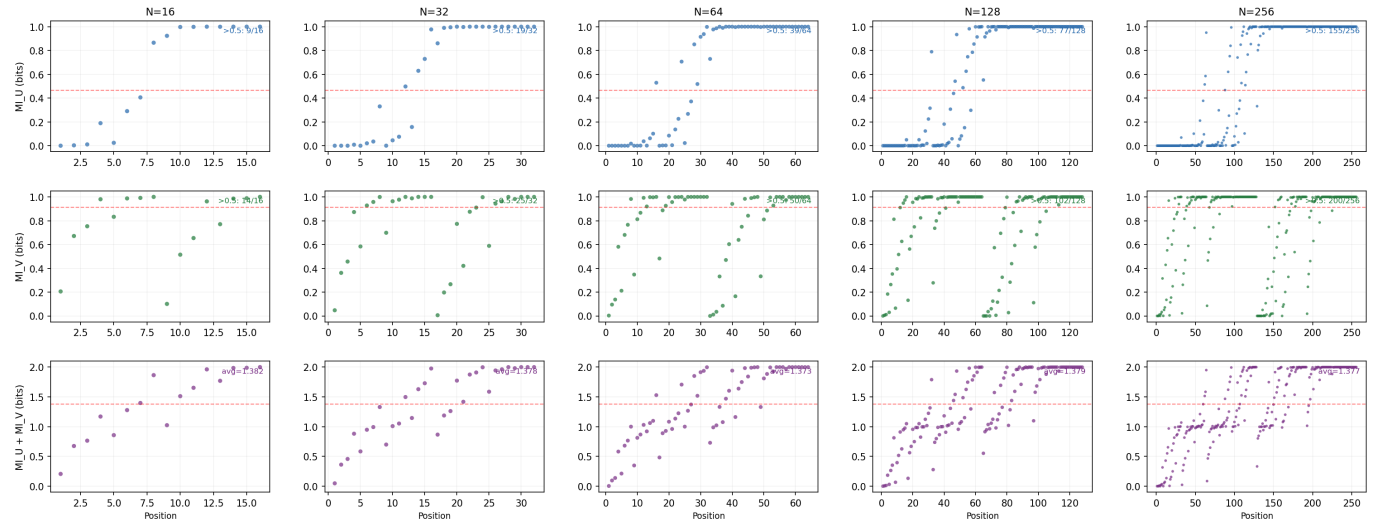
- $\text{MI}_i \in [0, 1]$: ranges from 0 (no information) to 1 (perfect prediction)
- **Conservation:** $\frac{1}{N} \sum_i \text{MI}_i^{(U)} = I(X; Z)$ and $\frac{1}{N} \sum_i \text{MI}_i^{(V)} = I(Y; Z|X)$
- **Polarization:** as $N \rightarrow \infty$, each $\text{MI}_i \rightarrow 0$ or 1

The red dashed line marks the channel capacity. The average MI across all positions equals the capacity.

GMAC (SNR = 6 dB): Class B

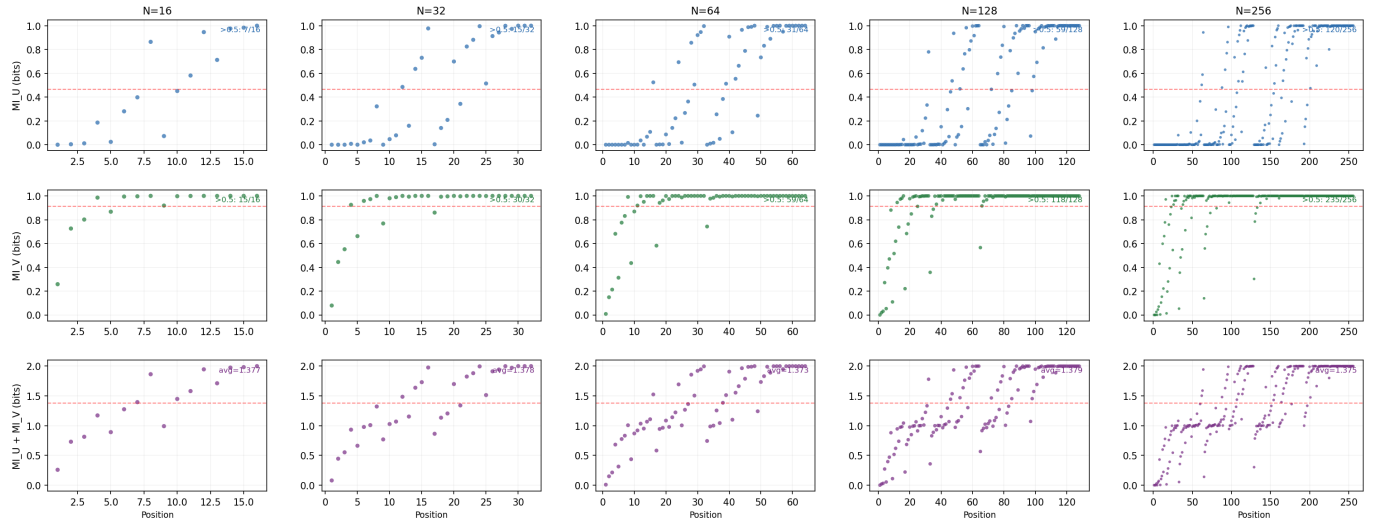
$$I(X;Z) = 0.465, I(Y;Z|X) = 0.912, I(X,Y;Z) = 1.376$$

GMAC (SNR=6dB) - Class B (interleaved)
Per-position MI from soft SC posteriors (genie-aided)
 $I(X;Z)=0.465$ $I(Y;Z|X)=0.912$ $I(X,Y;Z)=1.376$ (red dashed)



GMAC (SNR = 6 dB): Class C

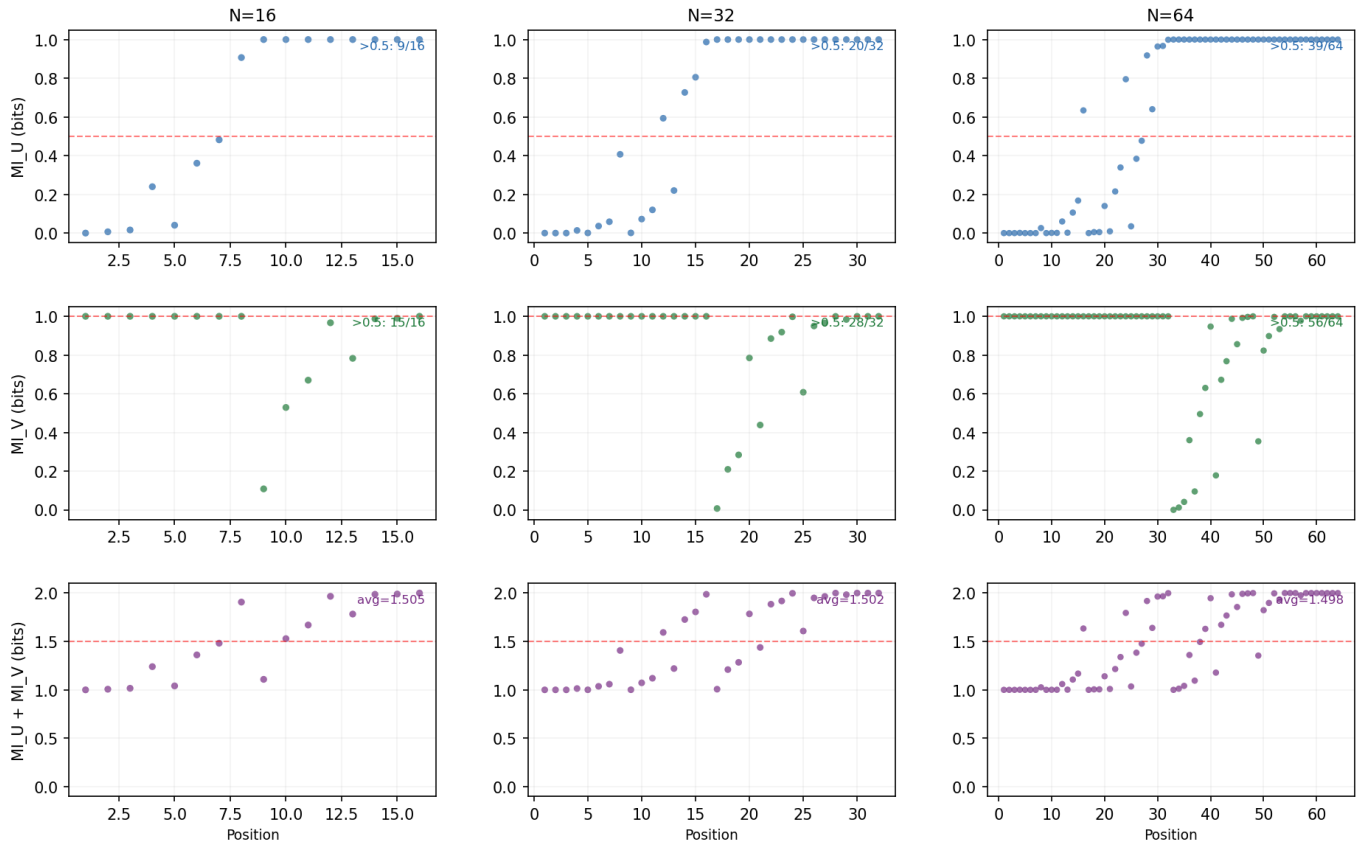
GMAC (SNR=6dB) - Class C (U first, then V)
Per-position MI from soft SC posteriors (genie-aided)
 $I(X;Z)=0.465$ $I(Y;Z|X)=0.912$ $I(X,Y;Z)=1.376$ (red dashed)



BEMAC: Class B

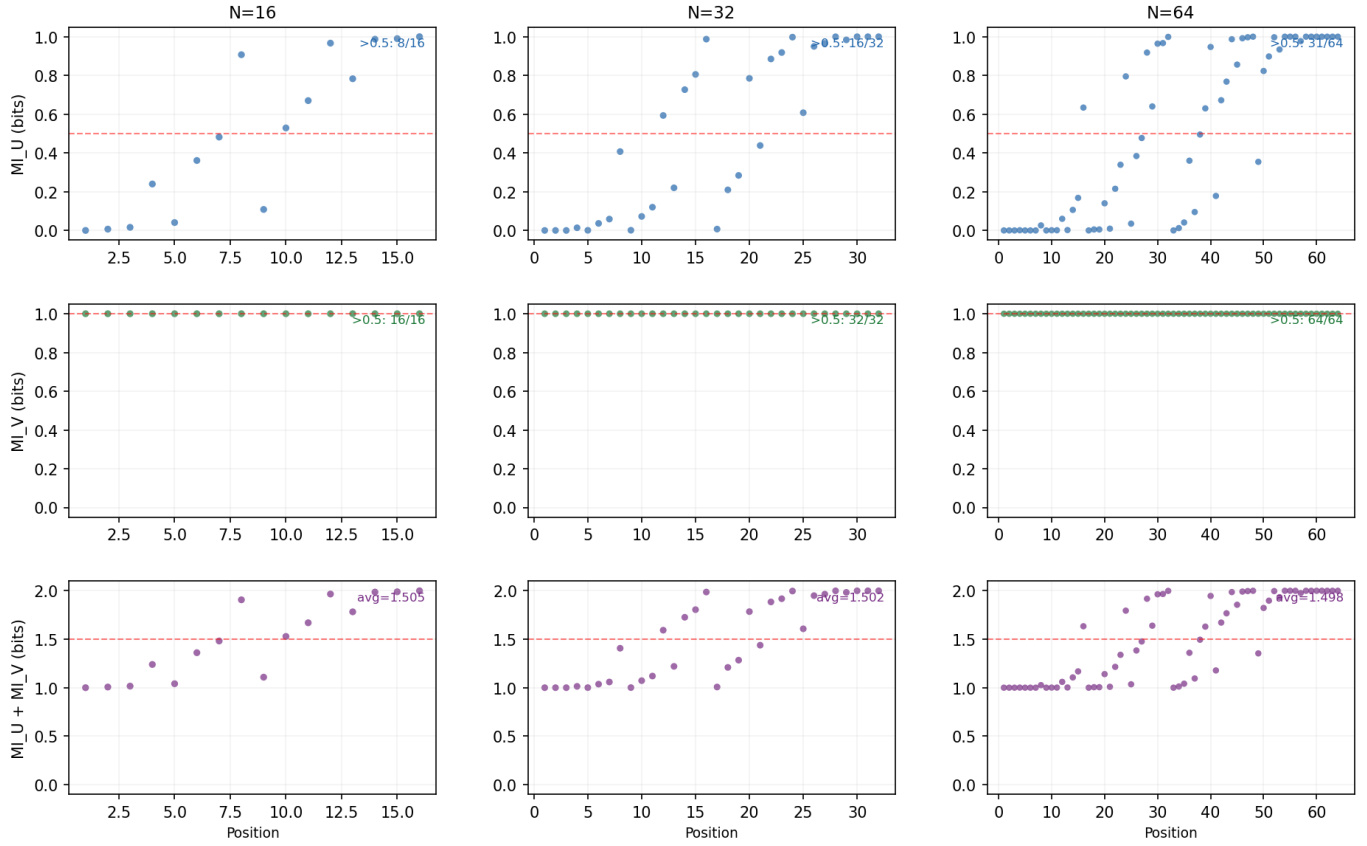
$$I(X; Z) = 0.500, I(Y; Z|X) = 1.000, I(X, Y; Z) = 1.500$$

BEMAC - Class B (interleaved)
Per-position MI from soft SC posteriors (genie-aided)
 $I(X; Z)=0.500$ $I(Y; Z|X)=1.000$ $I(X, Y; Z)=1.500$ (red dashed)



BEMAC: Class C

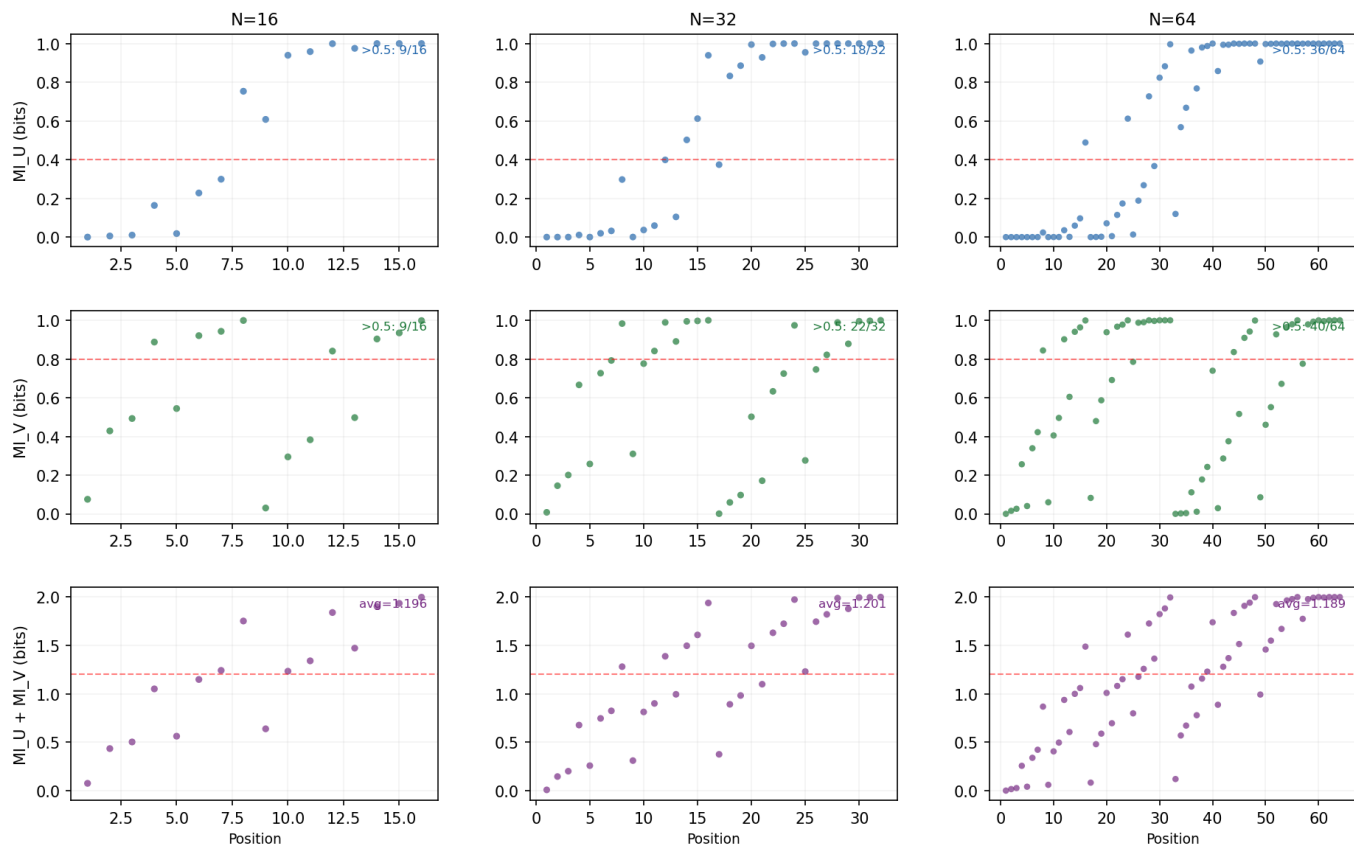
BEMAC - Class C (U first, then V)
 Per-position MI from soft SC posteriors (genie-aided)
 $I(X;Z)=0.500$ $I(Y;Z|X)=1.000$ $I(X,Y;Z)=1.500$ (red dashed)



ABNMAC: Class B

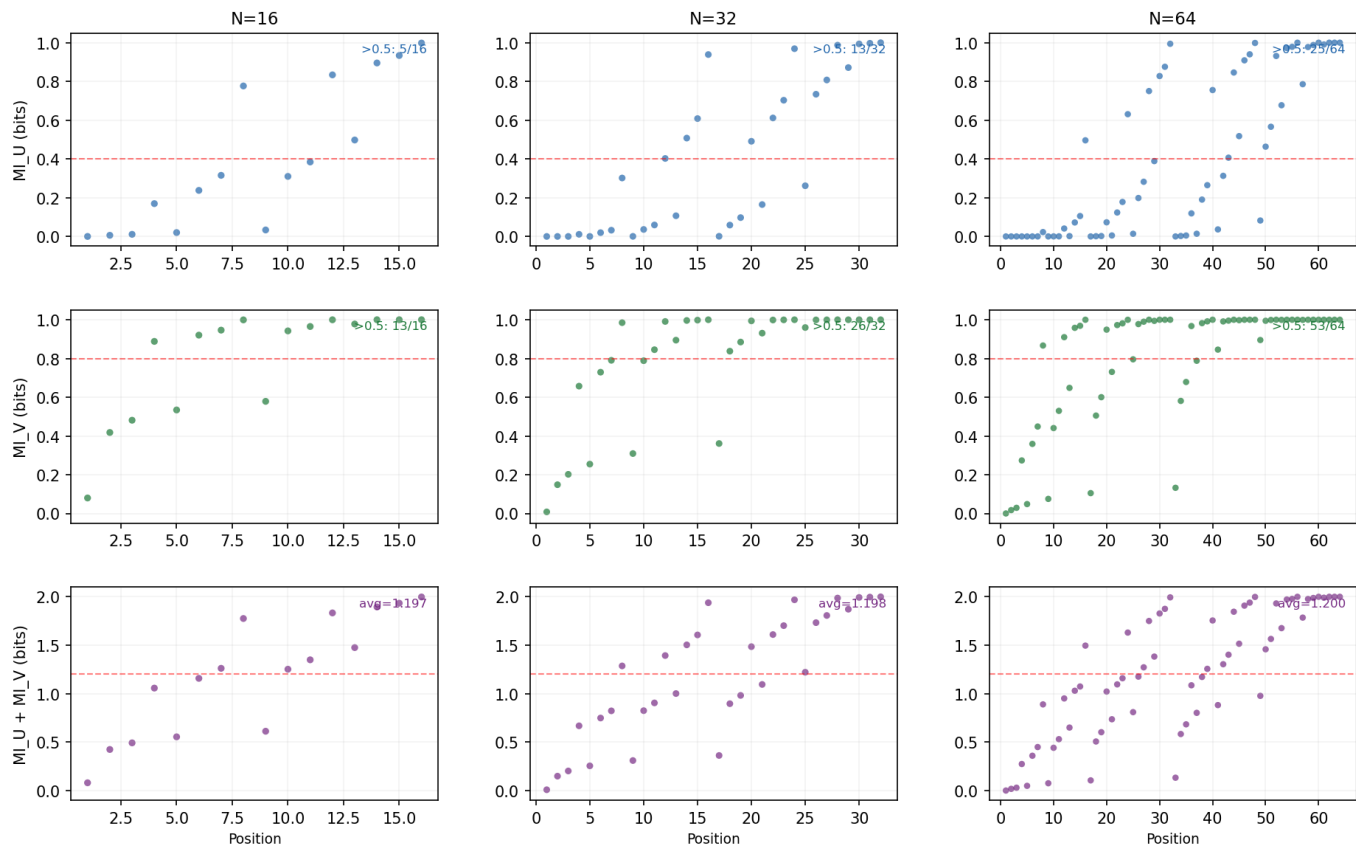
$$I(X; Z) = 0.400, I(Y; Z|X) = 0.800, I(X, Y; Z) = 1.200$$

ABNMAC - Class B (interleaved)
Per-position MI from soft SC posteriors (genie-aided)
 $I(X; Z)=0.400$ $I(Y; Z|X)=0.800$ $I(X, Y; Z)=1.200$ (red dashed)



ABNMAC: Class C

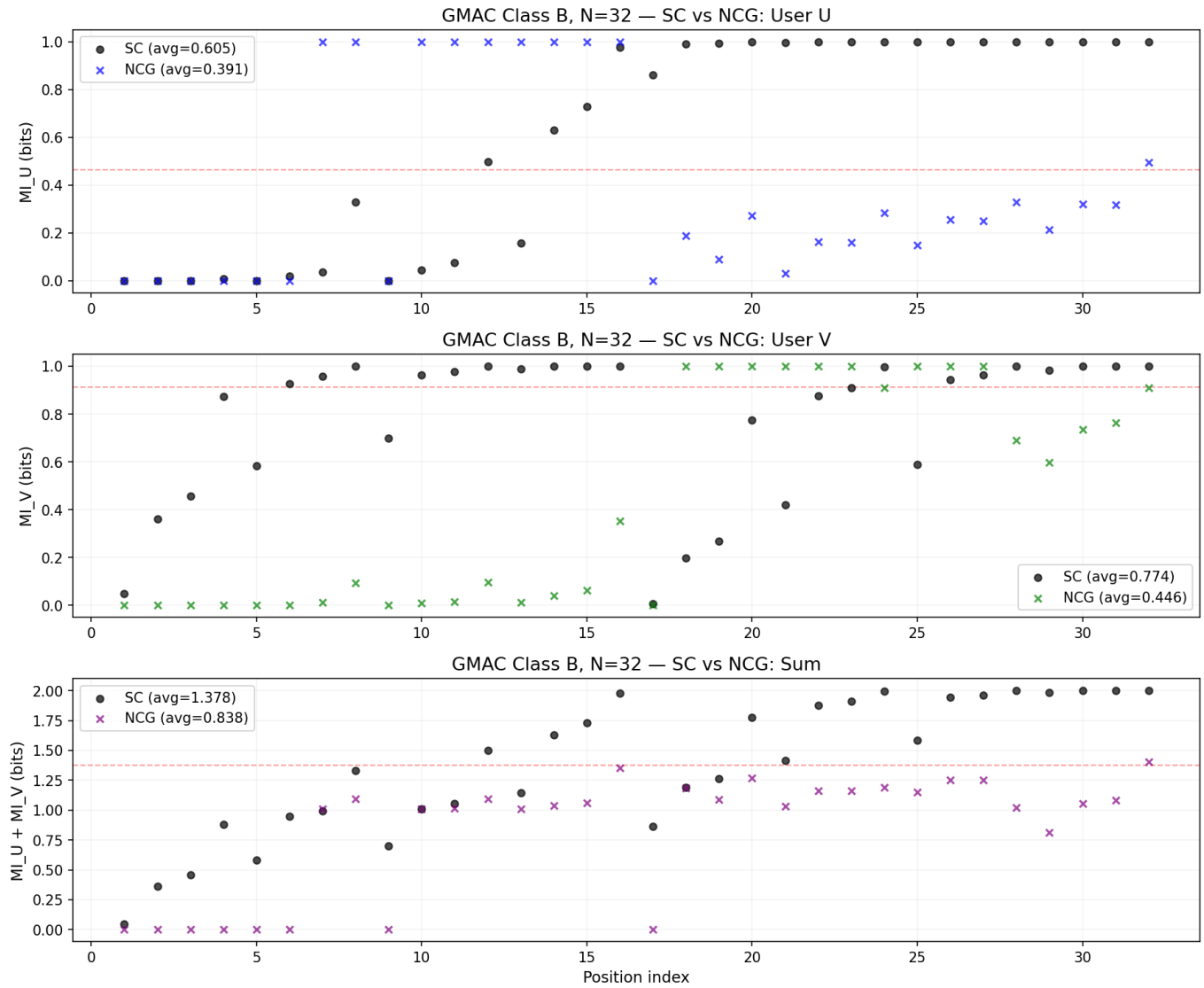
ABNMAC - Class C (U first, then V)
 Per-position MI from soft SC posteriors (genie-aided)
 $I(X;Z)=0.400$ $I(Y;Z|X)=0.800$ $I(X,Y;Z)=1.200$ (red dashed)



SC vs NCG Decoder: GMAC Class B

The NCG (Neural Computational Graph) decoder uses learned CalcLeft/CalcRight/CalcParent operations with teacher-forced sequential training. Black dots = SC analytical, colored crosses = NCG neural.

N=32



N=64

